

Computational Statistics - STAT 575 - HW #2

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Problem 1

Derive the E-M algorithm for right-censored normal data with known variance, say $\sigma^2 = 1$. Consider Y_i 's that are i.i.d. from a $N(\theta, 1)$, $i = 1, 2, \dots, n$. We observe $(x_1, \dots, x_n)$ and $(\delta_1, \dots, \delta_n)$, where $x_i = \min(y_i, c)$, and $\delta_i = I(y_i < c)$. Let C be the total number of censored (incomplete) observations. We denote the missing data as $\{Z_i : \delta_i = 0\}$.

Part (a)

Derive the complete log-likelihood, $l(\theta|Y)$.

The unobserved random variates $\{Y_i\}$ are i.i.d. normally distributed,

$$Y_i \sim f_{Y_i}(y|\theta)$$

where

$$f_{Y_i}(y|\theta) = (2\pi)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(y - \theta)^2\right).$$

The likelihood function is therefore

$$L(\theta|\{y_i\}) = \prod_{i=1}^n (2\pi)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(y_i - \theta)^2\right) \quad (1)$$

$$= (2\pi)^{-\frac{n}{2}} \exp\left(-\sum_{i=1}^n \frac{1}{2}(y_i - \theta)^2\right). \quad (2)$$

Taking the logarithm of L ,

$$\ell(\theta|\{y_i\}) = \log L(\theta|\{y_i\}) \quad (3)$$

$$= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (y_i - \theta)^2 \quad (4)$$

$$= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n y_i^2 + \theta \sum_{i=1}^n y_i - \frac{n}{2} \theta^2. \quad (5)$$

Anticipating that we will be maximizing the complete log-likelihood with respect to θ , we put any terms that are not a function of θ into k , obtaining the result

$$\ell(\theta|\{y_i\}) = k + \theta \sum_{i=1}^n y_i - \frac{n}{2} \theta^2.$$

Part (b)

Show the conditional expectation

$$E(Y|x, \delta = 1, \theta^{(t)}) = x$$

and

$$E(Y|x, \delta = 0, \theta^{(t)}) = E(Y|Y > x) = \theta^{(t)} + \frac{\phi(x - \mu)}{1 - \Phi(x - \mu)}$$

where ϕ and Φ are pdf and cdf of standard normal.

The distribution of Y given $\delta = 1$, is uncensored and therefore it is given that Y realized the value x . Since the expectation of a constant x is x , that means $E(Y|Y = x) = x$.

If $\delta = 0$, Y is censored, i.e., $Y > x$. To take its expectation, we first need to derive the conditional distribution of Y given $Y > x$ and $\theta^{(t)}$.

The probability $\Pr(Y \leq y|Y > x)$ is given by

$$\Pr(Y \leq y|Y > x) = \Pr(x < Y \leq y) / \Pr(Y > x)$$

which may be rewritten as

$$\Pr(Y \leq y|Y > x) = \frac{F_Y(y|\theta^{(t)}) - F_Y(x|\theta^{(t)})}{1 - F_Y(x|\theta^{(t)})}.$$

where $F_{Y|\theta^{(t)}}$ is the cdf of the normal distribution with $\sigma = 1$ and $\mu = \theta^{(t)}$.

We may rewrite $F_{Y|\theta^{(t)}}$ in terms of the standard normal,

$$F_Y(y|\theta^{(t)}) = \Phi(y - \theta^{(t)}),$$

and thus we may rewrite the conditional distribution of $Y|Y > x$ as

$$\Pr(Y \leq y|Y > x) = \frac{\Phi(y - \theta^{(t)}) - \Phi(x - \theta^{(t)})}{1 - \Phi(x - \theta^{(t)})}$$

and thus after further simplifying, we obtain the cdf of $Y|x$,

$$F_{Y|x}(y|\theta^{(t)}) = 1 - \frac{1 - \Phi(y - \theta^{(t)})}{1 - \Phi(x - \theta^{(t)})}$$

which has a density given by

$$f_Y(y|x, \theta^{(t)}) = \frac{\phi(y - \theta^{(t)})}{1 - \Phi(x - \theta^{(t)})} I(y > x).$$

The expectation of $Y|(x, \theta^{(t)})$ is given by

$$E(Y|x, \theta^{(t)}) = \int_x^\infty y f_Y(y|x, \theta^{(t)}) dy \tag{6}$$

$$= \int_x^\infty y \left(\frac{\phi(y - \theta^{(t)})}{1 - \Phi(x - \theta^{(t)})} \right) dy \tag{7}$$

$$= \frac{1}{1 - \Phi(x - \theta^{(t)})} \int_x^\infty y \phi(y - \theta^{(t)}) dy. \tag{8}$$

Analytically, this is a tricky integration problem. Certainly, it would be trivial to numerically integrate this to obtain a solution, but we seek a closed-form solution.

I searched online, and discovered an interesting way to tackle this integration problem.

Let f and F respectively denote the pdf and cdf of the normally distributed Y . Then,

$$\frac{df}{dy} = -(y - \theta)f(y)$$

and

$$\int_a^b \frac{df}{dy} dy = f(b) - f(a).$$

Then,

$$E(Y|x, \theta^{(t)}) = \frac{1}{1 - F(x)} \int_x^\infty yf(y)dy \quad (9)$$

$$= -\frac{1}{1 - F(x)} \int_x^\infty -(y - \theta^{(t)})f(y)dy + \frac{\theta^{(t)}}{1 - F(x)} \int_x^\infty f(y)dy \quad (10)$$

$$= -\frac{1}{1 - F(x)} \int_x^\infty \frac{df}{dy} dy + \frac{\theta^{(t)}}{1 - F(x)} (1 - F(x)) \quad (11)$$

$$= -\frac{1}{1 - F(x)} (f(\infty) - f(x)) + \theta^{(t)} \quad (12)$$

$$= \frac{f(x)}{1 - F(x)} + \theta^{(t)}. \quad (13)$$

We may rewrite the last line as

$$E(Y|x, \theta^{(t)}) = \theta^{(t)} + \frac{\phi(x - \theta^{(t)})}{1 - \Phi(x - \theta^{(t)})}.$$

Part (c)

Derive the E -step and M -step using parts (a) and (b). Give the updating equation.

E -step

The E -step entails taking the conditional expectation of the complete log-likelihood function $\ell(\theta|\{Y_i\})$ given the observed data $\{x_i\}$ and $\{\delta_i\}$.

$$Q(\theta|\theta^{(t)}) = E_{Y_i|x_i, \delta_i}(\ell(\theta|\{Y_i\})) \quad (14)$$

$$= E_{Y_i|x_i, \delta_i} \left(k + \theta \sum_{i=1}^n Y_i - \frac{n}{2} \theta^2 \right) \quad (15)$$

$$= k - \frac{n}{2} \theta^2 + \theta \sum_{i=1}^n E_{Y_i|x_i, \delta_i}(Y_i). \quad (16)$$

We have already solved the expectation of Y_i given x_i and δ_i . We rewrite Q by substituting $E(Y_i|x_i, \delta_i)$ with its previously found solution,

$$Q(\theta|\theta^{(t)}) = k - \frac{n}{2} \theta^2 + \theta \sum_{i=1}^n \delta_i x_i + (1 - \delta_i) \left(\theta^{(t)} + \frac{\phi(x_i - \theta^{(t)})}{1 - \Phi(x_i - \theta^{(t)})} \right).$$

Letting $C = \sum_{i=1}^n (1 - \delta_i)$, $R := \sum_{i=1}^n \delta_i x_i$, and separating out all terms that are independent of $\theta^{(t)}$,

$$Q(\theta|\theta^{(t)}) = k - \frac{n}{2} \theta^2 + C\theta\theta^{(t)} + R\theta + \theta \sum_{i=1}^n \frac{(1 - \delta_i)\phi(x_i - \theta^{(t)})}{1 - \Phi(x_i - \theta^{(t)})}.$$

M-step

We wish to solve

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta | \theta^{(t)}).$$

by solving

$$\left. \frac{dQ(\theta | \theta^{(t)})}{d\theta} \right|_{\theta=\theta^{(t+1)}} = 0,$$

which may be written as

$$-n\theta^{(t+1)} + C\theta^{(t)} + R + \sum_{i=1}^n \frac{(1 - \delta_i)\phi(x_i - \theta^{(t)})}{1 - \Phi(x_i - \theta^{(t)})} = 0.$$

Solving for $\theta^{(t+1)}$ obtains the updating equation

$$\theta^{(t+1)} = \frac{R}{n} + \frac{C}{n}\theta^{(t)} + \frac{1}{n} \sum_{i=1}^n \frac{(1 - \delta_i)\phi(x_i - \theta^{(t)})}{1 - \Phi(x_i - \theta^{(t)})}.$$

where

$$R := \sum_{i=1}^n \delta_i x_i$$

and

$$C := \sum_{i=1}^n (1 - \delta_i).$$

Part (d)

Use your algorithm on the V.A. data to find the MLE of μ . Take the log of the event times first and standardize by sample standard deviation. You may simply use the censored data sample mean as your starting value.

In the following R code, we implement the updating equation derived in the previous step. We encapsulate the procedure into a function that takes its arguments in the form of a censored set, uncensored set, starting value ($\theta^{(1)}$), and an ϵ value to control stopping condition.

```
# assuming the uncensored and censored data are distributed normally,
# we use the EM algorithm to derive an estimator given censored and uncensored
# data.
mean_normal_censored_estimator_em <- function(uncensored,censored,theta,eps=1e-6,debug=T)
{
  dev <- sd(log(c(uncensored,censored)))
  censored <- log(censored) / dev
  uncensored <- log(uncensored) / dev
  theta <- log(theta) / dev

  n <- length(censored) + length(uncensored)
  C <- length(censored)
  R <- sum(uncensored)

  s <- function(theta)
  {
    sum <- 0
    for (i in 1:C)
    {
```

```

    num <- dnorm(censored[i],mean=theta,sd=1)
    denom <- 1-pnorm(censored[i],mean=theta,sd=1)
    sum <- sum + (num / denom)
  }
  sum
}

i <- 1
repeat
{
  theta.new <- R/n + C/n * theta + (1/n)*s(theta)
  if (debug==T) { cat("theta[", i, "] =",theta," ", theta["", i+1, "] =",theta.new,"\n") }
  if (abs(theta.new - theta) < eps)
  {
    theta <- theta.new * dev
    theta <- exp(theta)
    return(theta)
  }
  i <- i + 1
  theta <- theta.new
}
}

```

We apply this procedure to the indicated data set.

```

library(MASS) # has VA data
VAs <- subset(VA,prior==0)
censored <- VAs$status == 0
censored_xs <- VAs[censored,c("stime")]
uncensored_xs <- VAs[!censored,c("stime")]

mu <- mean(uncensored_xs)
cat("mean of the uncensored sample is ", mu, ".")

## mean of the uncensored sample is 112.1648 .

sol <- mean_normal_censored_estimator_em(uncensored_xs,censored_xs,mu)

## theta[ 1 ] = 3.857928 , theta[ 2 ] = 3.424258
## theta[ 2 ] = 3.424258 , theta[ 3 ] = 3.415443
## theta[ 3 ] = 3.415443 , theta[ 4 ] = 3.415286
## theta[ 4 ] = 3.415286 , theta[ 5 ] = 3.415283
## theta[ 5 ] = 3.415283 , theta[ 6 ] = 3.415283

sol

## [1] 65.2625

```

We see that our estimate of θ is $\hat{\theta} = 65.2624985$. (The θ before transforming it to the appropriate scale was 3.415283.)

This mean is somewhat lower than anticipated, which makes me suspect something is wrong with my updating equation. If I have the time, I will revisit it.

Problem 2

Part (a)

There are $N = 1500$ gay men in the survey sample where X_i denotes the i -th persons response to the number of risky sexual encounters he had in the previous 30 days. Thus, we observe a sample $\vec{X} = (X_1, X_2, \dots, X_N)$.

We assume there are 3 groups in the population, denoted by $z = 1$, $t = 2$, and $p = 3$. Group 1 members report 0 risky sexual encounters regardless of the truth where the probability of being a member of group 1 is denoted by α ,

Group 2 members accurately report risky sexual encounters and represent typical behavior where the probability of being a member of group 2 is denoted by β . We assume this group's number of sexual encounters follows a poisson with mean μ .

Group 3 members accurately report risky sexual encounters and represent high-risk behavior where the probability of being a member of group 3 is $\gamma = 1 - \alpha - \beta$. We assume this group's number of sexual encounters follows a poisson with mean λ .

This represents a finite mixture model with a pdf

$$X_i \sim f(x|\vec{\theta}) = \alpha I(x=0) + \beta \text{POI}(x|\mu) + (1 - \alpha - \beta) \text{POI}(x|\lambda)$$

with a parameter vector

$$\vec{\theta} = (\alpha, \beta, \mu, \lambda)'$$

Let the uncertain group that the i -th person belongs to be denoted by Z_i . If we observe group membership data, $X_i|Z_i = z_i$, then

$$X_i|Z_i = 1 \sim I(x=0), \tag{17}$$

$$X_i|Z_i = 2 \sim \text{POI}(\mu), \tag{18}$$

$$X_i|Z_i = 3 \sim \text{POI}(\lambda), \tag{19}$$

where

$$Z_i \sim f_{Z_i}(z_i|\vec{\theta}) = \Pr(Z_i = z_i) = \begin{cases} \alpha & z_i = 1, \\ \beta & z_i = 2, \\ \gamma = 1 - \alpha - \beta & z_i = 3, \end{cases}$$

and thus

$$f_{X_i, Z_i}(x_i, z_i|\vec{\theta}) = \alpha I(z_i = 1) + \beta \text{POI}(\mu) I(z_i = 2) + (1 - \alpha - \beta) \text{POI}(\lambda) I(z_i = 3).$$

The *complete* likelihood function is thus given by

$$\mathcal{L}(\vec{\theta}|\vec{X}, \vec{Z}) = \prod_{i=1}^N f_{X_i, Z_i}(x_i, z_i|\vec{\theta}),$$

which may be rewritten as

$$\mathcal{L}(\vec{\theta}|\vec{X}, \vec{Z}) = \left(\prod_{\{i|z_i=1\}} \alpha I(x_i=0) \right) \left(\prod_{\{i|z_i=2\}} \beta \frac{\mu^{x_i} e^{-\mu}}{x_i!} \right) \left(\prod_{\{i|z_i=3\}} \gamma \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \right).$$

We wish to rewrite this so that the data is explicitly represented. First, we do the transformation

$$\mathcal{L}(\vec{\theta}|\vec{X}, \vec{Z}) = \left(\prod_{\{i|z_i=1, x_i=0\}} \alpha \right) \prod_{k=0}^{16} \left(\prod_{\{i|z_i=2, x_i=k\}} \beta \frac{\mu^k e^{-\mu}}{k!} \right) \prod_{k=0}^{16} \left(\prod_{\{i|z_i=3, x_i=k\}} \gamma \frac{\lambda^k e^{-\lambda}}{k!} \right).$$

We let $n_{a,b}$ denote the (unobserved) cardinality of $\{i|z_i = a, x_i = b\}$, thus

$$\mathcal{L}(\vec{\theta}|\{n_{j,k}\}) = \alpha^{n_{1,0}} \prod_{k=0}^{16} \beta^{n_{2,k}} \frac{\mu^{kn_{2,k}} e^{-\mu n_{2,k}}}{(k!)^{n_{2,k}}} \prod_{k=0}^{16} \gamma^{n_{3,k}} \frac{\lambda^{kn_{3,k}} e^{-\lambda n_{3,k}}}{(k!)^{n_{3,k}}}$$

is the complete likelihood. The complete log-likelihood is thus

$$\ell(\vec{\theta}|\{n_{j,k}\}) = n_{1,0} \log \alpha + \sum_{k=0}^{16} \log \left(\beta^{n_{2,k}} \frac{\mu^{kn_{2,k}} e^{-\mu n_{2,k}}}{(k!)^{n_{2,k}}} \right) + \sum_{k=0}^{16} \log \left(\gamma^{n_{3,k}} \frac{\lambda^{kn_{3,k}} e^{-\lambda n_{3,k}}}{(k!)^{n_{3,k}}} \right)$$

which simplifies to

$$\begin{aligned} \ell(\vec{\theta}|\{n_{j,k}\}) = n_{1,0} \log \alpha + \sum_{k=0}^{16} n_{2,k} (\log \beta + k \log \mu - \mu - \log k!) + \\ n_{3,k} (\log \gamma + k \log \lambda - \lambda - \log k!). \end{aligned} \quad (20)$$

Anticipating taking $\frac{d\ell}{d\vec{\theta}}$ to solve for the maximum of the log-likelihood, we remove any terms that are not a function of $\vec{\theta}$, resulting in the kernel

$$\ell(\vec{\theta}|\{n_{j,k}\}) = n_{1,0} \log \alpha + \sum_{k=0}^{16} \{n_{2,k} (\log \beta + k \log \mu - \mu) + n_{3,k} (\log \gamma + k \log \lambda - \lambda)\}.$$

E-step

The conditional expectation to solve in the EM algorithm is given by

$$Q(\vec{\theta}|\vec{\theta}^{(t)}) = E(\ell(\vec{\theta}))$$

where $\{n_{k,j}\}$ are random and $\{n_j\}$ and $\vec{\theta}^{(t)}$ are given. We rewrite this as

$$Q(\vec{\theta}|\vec{\theta}^{(t)}) = E \left(n_{1,0} \log \alpha + \sum_{k=0}^{16} \{n_{2,k} (\log \beta + k \log \mu - \mu) + n_{3,k} (\log \gamma + k \log \lambda - \lambda)\} \right).$$

Using the linearity of expectations, we rewrite the above to

$$Q(\vec{\theta}|\vec{\theta}^{(t)}) = E(n_{1,0}) \log \alpha + \sum_{k=0}^{16} \{E(n_{2,k}) (\log \beta + k \log \mu - \mu) + E(n_{3,k}) (\log \gamma + k \log \lambda - \lambda)\}$$

given $\{n_j\}$ and $\theta^{(t)}$.

Consider $E(n_{2,k}|\{n_j\}, \theta^{(t)})$. To solve this expectation, we must first derive the distribution of $n_{2,k}$.

Suppose $x_j = k$, then probability that the j -th person belongs to group 2 is given by

$$\Pr(Z_j = 2|x_j = k) = \Pr(Z_j = 2) \Pr(x_j = k|Z_j = 2) / \Pr(x_j = k).$$

We note that $\Pr(x_j = k)$ is equivalent to $\pi_k(\vec{\theta})$, $\Pr(Z_j = 2)$ is the definition of β , and $\Pr(x_j = k|Z_j = 2)$ is $f_{X_j|Z_j}(k|Z_j = 2) = \text{POI}(k|\mu)$.

Making the substitutions yields the result

$$t_k(\vec{\theta}) = \Pr(Z_j = 2|x_j = k) = \beta \text{POI}(k|\mu) / \pi_k(\vec{\theta}).$$

Assuming $\{X_i\}$ are i.i.d., observe that $k \neq 0$, the distribution of $n_{2,k}$ given $n_k, \theta^{(t)}$ is binomial distributed with a probability of success $t_k(\vec{\theta}^{(t)})$. Thus,

$$\mathbb{E}(n_{2,k}) = n_k t_k(\vec{\theta}^{(t)}).$$

The same logic holds for $n_{3,k}$ and $n_{1,0}$, and thus

$$\mathbb{E}(n_{3,k}) = n_k p_k(\vec{\theta}^{(t)})$$

and

$$\mathbb{E}(n_{1,0}) = n_0 z_0(\vec{\theta}^{(t)}),$$

which means

$$Q(\vec{\theta}|\vec{\theta}^{(t)}) = n_0 z_0(\vec{\theta}^{(t)}) \log \alpha + \sum_{k=0}^{16} \left\{ n_k t_k(\vec{\theta}^{(t)}) (\log \beta + k \log \mu - \mu) + n_k p_k(\vec{\theta}^{(t)}) (\log \gamma + k \log \lambda - \lambda) \right\}$$

M-step

We wish to solve

$$\vec{\theta}^{(t+1)} = \arg \max_{\vec{\theta}} Q(\vec{\theta}|\vec{\theta}^{(t)}).$$

by solving

$$\nabla Q(\vec{\theta}|\vec{\theta}^{(t)}) \Big|_{\vec{\theta}=\vec{\theta}^{(t+1)}} = \vec{0}.$$

We use the Lagrangian to impose the restriction $\alpha + \beta + \gamma = 1$, thus we seek to perform the constrained maximization of

$$Q_l(\vec{\theta}, c|\vec{\theta}^{(t)}) = Q(\vec{\theta}|\vec{\theta}^{(t)}) + c(1 - \alpha - \beta - \gamma).$$

Thus, when we solve for α ,

$$\frac{\partial Q_l}{\partial \alpha} = \frac{n_0 z_0(\theta^{(t)})}{\alpha} - c = 0,$$

we get the result

$$\alpha^{(t+1)} = \frac{1}{c} n_0 z_0(\theta^{(t)}).$$

Similar results hold for β and γ , obtaining

$$\beta^{(t+1)} = \frac{1}{c} \sum_{k=0}^{16} n_k t_k(\theta^{(t)}).$$

and

$$\gamma^{(t+1)} = \frac{1}{c} \sum_{k=0}^{16} n_k p_k(\theta^{(t)}).$$

This does not look too promising until we realize that

$$n_0 z_0(\theta^{(t)}) + \sum_{k=0}^{16} n_k t_k(\theta^{(t)}) + \sum_{k=0}^{16} n_k p_k(\theta^{(t)}) = N.$$

Thus, $c(\alpha^{(t)} + \beta^{(t)} + \gamma^{(t)}) = N$, which means $c = N$ since $\alpha^{(t)} + \beta^{(t)} + \gamma^{(t)} = 1$. Making this substitution obtains the result

$$\alpha^{(t+1)} = \frac{1}{N} n_0 z_0(\theta^{(t)}) \quad (21)$$

$$\beta^{(t+1)} = \frac{1}{N} \sum_{k=0}^{16} n_k t_k(\theta^{(t)}) \quad (22)$$

$$\gamma^{(t+1)} = \frac{1}{N} \sum_{k=0}^{16} n_k p_k(\theta^{(t)}). \quad (23)$$

Solving an estimator for μ at iteration $(t+1)$,

$$\left. \frac{\partial Q_l}{\partial \mu} \right|_{\mu=\mu^{(t+1)}} = 0 \quad (24)$$

$$\sum_{k=0}^{16} n_k t_k(\theta^{(t)}) (k/\mu^{(t+1)} - 1) = 0 \quad (25)$$

$$\frac{1}{\mu^{(t+1)}} \sum_{k=0}^{16} n_k t_k(\theta^{(t)}) k = \sum_{k=0}^{16} n_k t_k(\theta^{(t)}) \quad (26)$$

$$\mu^{(t+1)} = \frac{\sum_{k=0}^{16} k n_k t_k(\theta^{(t)})}{\sum_{k=0}^{16} n_k t_k(\theta^{(t)})}. \quad (27)$$

The same derivation essentially follows for λ , and thus

$$\lambda^{(t+1)} = \frac{\sum_{k=0}^{16} k n_k p_k(\theta^{(t)})}{\sum_{k=0}^{16} n_k p_k(\theta^{(t)})}.$$

Part (b)

Estimate the parameters of the model, using the observed data.

```
# we observe n = (n0,n1,...,n16)
ns <- c(379,299,222,145,109,95,73,59,45,30,24,12,4,2,0,1,1)
N <- sum(ns)

# theta := (alpha, beta, mu, lambda)'
# note that there is an implicit parameter gamma s.t.
# alpha + beta + gamma = 1
# the initial value assumes each category z, t, or p
# is equally probable, and so we let
# (alpha^(0),beta^(0)) = (1/3,1/3)
# and mu^(0) and lambda^(0) are just arbitrarily chosen to be 2 and 3,
# with the insight that group 3 is more risky than group 2.
theta <- c(1/3,1/3,2,3)

# theta := (alpha, beta, mu, lambda)
Pi <- function(i,theta)
{
  res <- 0
  if (i == 0)
    res <- theta[1]
```

```

res <- res + theta[2] * theta[3]^i * exp(-theta[3])
res <- res + (1 - theta[1] - theta[2]) * theta[4]^i * exp(-theta[4])
res
}

z0 <- function(theta)
{
  theta[1] / Pi(0,theta)
}

t <- function(i,theta)
{
  theta[2] * theta[3]^i * exp(-theta[3]) / Pi(i,theta)
}

p <- function(i,theta)
{
  (1-theta[1] - theta[2]) * theta[4]^i * exp(-theta[4]) / Pi(i,theta)
}

# update algorithm, based on EM algorithm
update <- function(theta,ns)
{
  # note: n0 := ns[1] instead of ns[0] since R does not use zero-based indexes
  alpha <- ns[1] * z0(theta) / N
  beta <- 0
  mu_num <- 0
  mu_denom <- 0

  lam_num <- 0
  lam_denom <- 0

  for (i in 0:16)
  {
    ti <- t(i,theta)
    pi <- p(i,theta)

    beta <- beta + ns[i+1] * ti

    mu_num <- mu_num + i * ns[i+1] * ti
    mu_denom <- mu_denom + ns[i+1] * ti

    lam_num <- lam_num + i * ns[i+1] * pi
    lam_denom <- lam_denom + ns[i+1] * pi
  }

  beta <- beta / N
  mu <- mu_num / mu_denom
  lam <- lam_num / lam_denom

  c(alpha,beta,mu,lam)
}

```

```

em <- function(theta,ns,steps=10000,debug=T)
{
  for(i in 1:steps)
  {
    theta = update(theta,ns)
    if (debug==T)
    {
      if (i %% 1000 == 0) { cat("iteration =",i," theta = (",theta,")'\n") }
    }
  }
  theta
}

# solution theta = (alpha, beta, mu, lambda)
sol <- em(theta,ns,10000,T)

## iteration = 1000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 2000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 3000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 4000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 5000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 6000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 7000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 8000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 9000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'
## iteration = 10000  theta = ( 0.1221661 0.5625419 1.467475 5.938889 )'

```

We see that the solution is 0.1221661,0.5625419,1.4674746,5.9388889.

Part (c)

Estimate the standard errors and pairwise correlations of your parameters, using any available method.

We have chosen to use the Bootstrap method.

```

# ns = (379,299,222,145,109,95,73,59,45,30,24,12,4,2,0,1,1)
# 379 responded 0 encounters
# 299 responded 1 encounters
# 222 responded 2 encounters
# ...
# 1 responded 16 encounters
#
# to resample, we resample from the data set that includes each
# persons response, as determined by ns.
data <- NULL
for (i in 1:length(ns))
{
  data <- append(data,rep((i-1),ns[i]))
}

make_into_counts <- function(data)
{
  ns <- NULL
  for (i in 0:16)
  {

```

```

    ni <- data[data == i]
    l <-length(ni)
    ns <- append(ns,l)
  }
  ns
}

m <- 1000 # bootstrap replicates
steps <- 500
theta.bs <- em(theta,ns,steps,F)
thetas <- rbind(theta.bs)
for (i in 2:m)
{
  indices <- sample(N,N,replace=T)
  resampled <- make_into_counts(data[indices])
  theta.bs <- em(theta,resampled,steps,F)
  thetas <- rbind(thetas,theta.bs)
  if (i %% 100 == 0) { cat("iteration", i, ": ", theta.bs, "\n") }
}

```

```

## iteration 100 :  0.1202674 0.5681112 1.519115 6.09385
## iteration 200 :  0.1345963 0.5588609 1.588654 6.207387
## iteration 300 :  0.09952392 0.565353 1.369021 5.52042
## iteration 400 :  0.1204714 0.5681574 1.45573 5.723874
## iteration 500 :  0.1240077 0.5468764 1.502662 5.83856
## iteration 600 :  0.1116724 0.5736862 1.60266 5.962054
## iteration 700 :  0.1161729 0.5769302 1.486408 6.129461
## iteration 800 :  0.1244822 0.575823 1.481806 6.09534
## iteration 900 :  0.08395545 0.5999121 1.401224 5.801542
## iteration 1000 : 0.1248928 0.5640767 1.444692 6.022184

```

```

cov.bs <- cov(thetas)
cor.bs <- cor(thetas)

```

The Bootstrap estimator of the covariance matrix is given by

```

##           [,1]           [,2]           [,3]           [,4]
## [1,]  0.0004284232 -0.0002202565  0.001656424 0.001417201
## [2,] -0.0002202565  0.0004590897 -0.000108595 0.001307151
## [3,]  0.0016564240 -0.0001085950  0.012243409 0.012747923
## [4,]  0.0014172014  0.0013071507  0.012747923 0.036646178

```

and the correlation matrix is given by

```

##           [,1]           [,2]           [,3]           [,4]
## [1,]  1.0000000 -0.49664198  0.72324230 0.3576685
## [2,] -0.4966420  1.00000000 -0.04580467 0.3186857
## [3,]  0.7232423 -0.04580467  1.00000000 0.6018301
## [4,]  0.3576685  0.31868567  0.60183012 1.0000000

```